# Phantom Uptime

Finding the rapid-charger failures that never reach the alert queue.

Agentic reliability copilot · Umut Özbağrıaçık · 2026

FICTIONAL OPERATOR

**Voltway Networks**

Synthetic pilot · 600 chargers · 150 sites · Midlands & North

**Synthetic scenario.** Voltway Networks is a fictional operator. All data, metrics and findings are generated, no real operator or customer data is used, and the prototype acts on no real system.

## THE PROBLEM

Rapid EV charging is one of the few sectors where reliability is at once a **regulated and published number** and a direct revenue line.

**A charger can look compliant to the reporting layer, healthy to the alerting system, and dead to the driver.**

Under the **UK Public Charge Point Regulations 2023**, a network's 50kW+ rapid public charge points are measured against a **99% average reliability requirement** across the calendar year, computed from charge point status data. A unit that keeps reporting operational statuses can look healthy in status-derived reporting while driver outcomes fail, and it stays out of an alert queue built from the fault log, however many drivers it turns away. That makes one failure both a **compliance exposure** and a **commercial blind spot**: lost sessions, lost kWh, site-partner friction and driver trust.

- **Reported reliability** is status-derived uptime, a share of *time*.
- **Delivered reliability** is the share of driver attempts that transferred energy, a share of *attempts*.
- Different denominators, so these are not two versions of one number.
- A silent failure runs **Available → Preparing → Available**: never Charging, never Faulted, nothing for the alert queue to work.

The core problem is the gap between reported and delivered reliability, and the fact that the alert queue may not contain the chargers doing the commercial damage.

### WHAT THE PILOT DATA SHOWS

| | | |
|---|---|---|
| **99.26%** REPORTED RELIABILITY | **95.29%** DELIVERED RELIABILITY | **3.96pts** THE GAP |
| **0** ALERT OVERLAP | **£12.6k** HIDDEN REVENUE AT RISK | **£708/day** ONGOING LEAKAGE |

**123 alerts across 106 devices. 26 high-confidence silent failures. Not one charger appears in both.** Three of those episodes are still running today.

## WORKED EXAMPLE

**CHG-0034 · Birmingham Motorway 006 · 2–4 September**

| | |
|---|---|
| Last successful charge | 1 Sept, 23:41 → 00:05 |
| Attempts since | 63 |
| Successful | 0 |
| Energy delivered | 0 kWh |
| Reported uptime | 100% |
| Faults · alerts raised | 0 · 0 |
| The 5 other chargers on site | 95.75% success |
| Revenue at risk · still running | £1,151 · £384/day |

**The first engineer goes here**, not because this unit has lost the most so far, but because it is the one still losing. No fault state or alert was ever triggered, so it never entered the operations queue at all.

## ASSUMPTIONS

- All operational data is synthetic; every finding is fictional and describes no real operator.
- The pilot covers 600 chargers across 150 sites in the Midlands and the North of England over one month, so it says nothing about seasonality or the wider estate.
- Reported reliability is modelled as uptime across non-fault operational statuses; delivered reliability counts an attempt as successful when energy was transferred. Which statuses count is the assumption most likely to differ in production.
- Revenue figures are estimates for operational prioritisation, not accounting-grade lost revenue, and assume a turned-away driver did not simply use the next unit.
- Nothing is dispatched, sent or ticketed without a person approving it.

## SOLUTION ARCHITECTURE

**Operational source data**
OCPP status · charging attempts · payments · faults · alerts

↓

**OLTP → OLAP transformation**
DuckDB SQL builds the analytical marts

↓

**Analytical marts**
reliability · silent failures · alert blind spot · commercial impact

↓

**Cube semantic layer**
governed metrics and definitions, travelling with every value

↓

**Approved tools**
reliability · blind spot · evidence · impact · action draft

↓

**Tool-calling agent**
investigates, explains the evidence, recommends the dispatch

↓

**Copilot interface**
dashboard · trace · evidence · human-approved action

The model does not query raw data, write SQL, or invent metric definitions. Cube owns the metrics; approved tools own the arithmetic; the model chooses which tool to call and how to argue the result.

## TRUST AND GOVERNANCE

- **Conservative detection.** High-confidence, actionable failures are preferred over a noisy queue: 26 of 35 seeded impairments found, no false positives.
- **Governed metrics.** Cube owns every definition; the model cannot invent a calculation or reach the tables underneath.
- **Trace validation.** Every important figure in an answer is matched in code to a tool output, on the server, so an API caller gets the same check as a viewer. Across the recorded set that is 116 figures, none unsourced. The same code runs in CI on every push, alongside the detector's own 28 checks.
- **Detection from data.** Silent failures are found in SQL, never read from the generator's own record of what it planted. A test enforces that.

## QUESTIONS FOR A REAL DEPLOYMENT

1. How is reliability calculated and published today, and **which charge point statuses are included in or excluded from uptime?** Whether Preparing counts materially affects this pattern and how it shows up in reported reliability.
2. Are **delivered outcomes** (attempts that transferred energy) measured anywhere today, or only uptime?
3. Which failure modes does the current alerting miss, and how do they surface: driver complaints, the site partner, a field visit?
4. How is **the first engineer** chosen today: severity, revenue, contractual commitment, or geography?
5. What evidence is needed before acting on a charger that reports itself healthy?
6. Who owns the reliability number commercially, and **what happens if it is restated downward?** Technical fix, political problem.

## DELIBERATELY OUT OF SCOPE

No live OCPP connection, no ticketing integration, no pricing engine, and no second model provider behind an abstraction that has one implementation. Known integration work was left out on purpose. The question worth prototyping was whether an agent can produce trusted, evidence-backed operational judgement.